

Candidate Number _____ Score _____

2019 Republic of China Biology Olympiad Selection Camp Practical Test QuestionsPractice **A** in the **4th** Examination Room

Plant Physiology Practical Exercises

Physiological manifestations of seed soaking (50 points)

I. Experimental Materials:			
Seed (mung bean) pretreatment			
Seeds in group A (re-drying): Soaked for 12 hours, air-dried and stored for one day, then soaked for another 2 hours before the experiment.			7
Seeds in group B (control group): Soaked for 2 hours before the experiment.			7
Seeds in group C (re-drying): 1.5 ml of amylase extract (in an ice bucket).			1 tube
Control group D: 1.5ml amylase extract (in an ice bucket) II.			1 tube
Experimental equipment and reagents:			
Experimental equipment	Quantity:	Drug solution	quantity
1. Tweezers	1 vial of	Iodine solution	10ml
2. Single-sided blade	One	2.04% starch solution	15ml
3. Micropipettes P1000	1 vial of	TTC solution (protected from	3.5ml
4. Micropipettes P200	light) 1 vial of	TTC solution 1 box of distilled	15ml
5. Micropipettes tip P1000 6.	water (deionized water) 1 box of	50% TCA	2ml
Micropipettes tip P200 7.	(protected from light) 1 ruler		
Strange pen	1 piece of	Petri dish A (including filter paper) (reference number)	
8. Cuvette 9. 1.5 ml	6. Petri dish B (containing starch and agar-agar) (Science Reference Number)	1 piece	
centrifuge tube 10. 15	6		
ml centrifuge tube	3		
11. 3 ml dropper	1 piece of shared equipment:		
12. Timer (do not set manually)	1 spectrophotometer (shared)		
13. Aluminum	1 centrifuge (shared)		
foil 14.	1 unit		
Calculator	1		
15. Adhesive tape 16. Gloves	1 pair		

Please note:

1. Please check if the candidate number on the question matches your candidate number.
2. The materials and equipment on the table will not be replenished once they are used up.
3. This exam paper (including the cover and the question paper) consists of 6 pages and must be returned in its entirety upon submission.
4. This exam includes two parts: Practice A and Practice B. The total time allotted for answering the questions is 90 minutes.
5. Please answer the questions on this paper. You may write your answers on the back of the paper, but please indicate the question number.

III. Experiment and Principle Explanation:

On Saturday morning, Xiaoming soaked the precious seeds that his teacher had worked so hard to obtain in water, preparing to...

Under the guidance of the teacher, I was conducting scientific experiments and research for the exhibition. After waiting a long time, I received an urgent call from my teacher, who said there was an unforeseen emergency.

The science exhibition experiment had to be postponed. Xiao Ming was stunned for a long time before quickly scooping out the seeds that had been soaking for several hours.

Air-dry and store. A few days later, the teacher finally had time to guide Xiaoming in his science fair experiment. Xiaoming took out the air-dried seeds and told...

Knowing the teacher's concerns, the teacher worried that the seeds' vigor would be affected. After thinking for a while, the teacher decided to guide Xiaoming in seed...

Physiological studies of infiltration were conducted using experimental methods such as respiratory activity (TTC), starch degradation capacity, and amylase activity.

An experiment was conducted to investigate and explore the effect of soaking and re-drying treatment on seed vigor. The teacher asked Xiaoming to soak mung bean seeds...

After soaking in water for 12 hours, the seeds were removed and air-dried naturally to form the re-drying treatment group. Separately, some completely untreated seeds were used as...

The control group and the seeds of both groups were soaked in water for 2 hours before the experiment. The following is Xiaoming's experiment.

Please understand and follow the principles and steps:

TTC Seed Viability Detection Method Experimental Principle

TTC is a commonly used method for rapid detection of seed viability; TTC stands for triphenyltetrazolium chloride (2,3,5-

TTC is an abbreviation for triphenyl tetrazolium chloride. The principle of its testing is that the oxidized form of TTC is colorless and soluble.

Sex, dehydrogenases within living cells use electrons and hydrogen atoms produced during respiration to reduce colorless TTC to

The red compound triphenylformazan, however, is absent in dead cells due to the lack of dehydrogenase.

The color change reaction can be used to test the intensity of respiration in different parts of the seed.

Experimental principle of starch degradation capacity

This experiment primarily observes the effect of amylase in mung bean seeds after soaking and then drying.

When seeds are soaked and saturated with water, the endosperm produces amylase to break down the starch in the embryo. Amylase is an enzyme...

(Protein) Its function is to break down large starch molecules into smaller carbohydrate molecules. Starch is composed of starch sugars.

A large molecular compound composed of amylose and amylopectin. The amylose component is composed of...

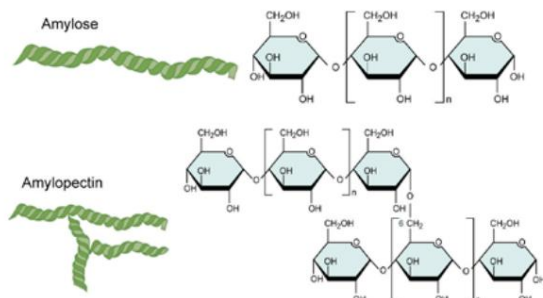
Glucose molecules are arranged in a linear $\alpha(1-4)$ chain; amylopectin, on the other hand, is composed of $\alpha(1-6)$ molecules containing phosphate groups.

Branched structure (as shown in the diagram below). Iodide ions (I_2) react with starch sugars to produce a blue color and with starch gum to produce a red color.

The purple color is called the iodine-starch reaction. The greater the fading of the iodine-starch color, the more starch has been degraded.

Starch sugar

Gum starch



Principle of α -Amylase Activity Assay

α -Amylase is a starch-degrading enzyme. The hydrolysis by amylase is unidirectional and irreversible.

The product is maltose. In the iodine-starch reaction, the compound formed by the iodine solution and starch...

The absorbance at 490 nm is the maximum and can be used for quantification. This experiment utilizes extracts from the tested seeds.

The enzyme solution reacts with starch solution, and the starch consumption is then measured using iodine solution to estimate the enzyme activity of the sample.

sex.

IV. Experimental Methods:

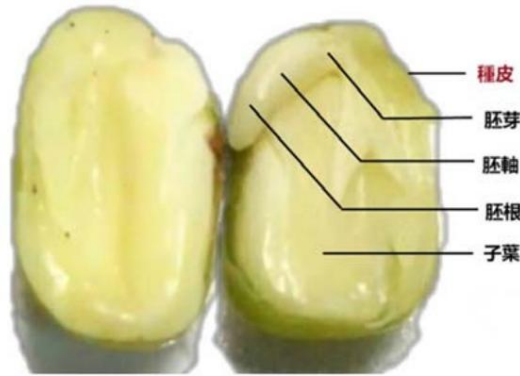
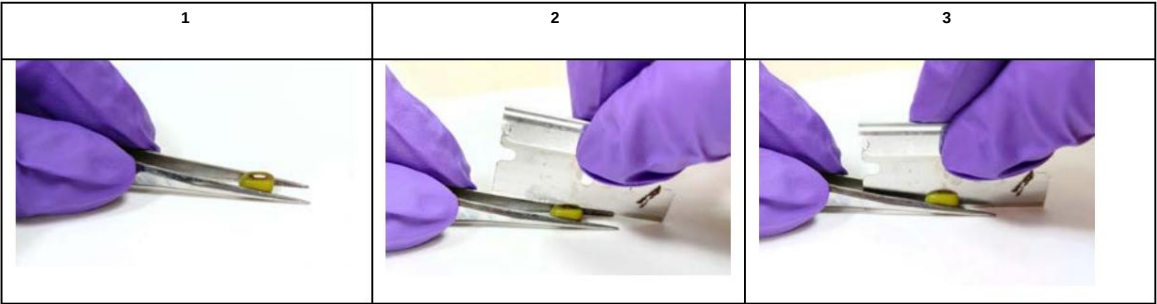


Figure 1. Longitudinal section of mung bean seeds after soaking for 12 hours.

Please select five seeds from each of the dry group and the control group, which have been soaked in water for 2 hours. Please follow the instructions below.

Images 1, 2, and 3 are cut in half respectively (as shown in Figure 1).



A. Respiratory activity assay: First, add 3.5 mL of TTC solution to petri dish A.

Place the seeds as shown in Figure 2, and arrange the 5 halved seeds in the grid according to the order of the squares (please place the same seed in the grid).

Place one half in petri dish A and the other half in the corresponding position in petri dish B. Cover petri dish A with aluminum foil.

After placing the paper (away from light), let it sit for 15 minutes. Once finished, raise your hand and ask the teaching assistant to take a photo and record the observed reaction results.

B. Starch degradation capacity:

- (1) Place petri dish B on the table shown in Figure 2, and place the cut surface of the halved seed flat on the starch-containing surface.

Place the carrageenan culture dish B into the grid shown in Figure 2 and let it stand for 15 minutes.

- (2) Using a 3ml dropper, add 3mL of iodine solution and shake left and right 5 times (do not splash). Let stand for 5 minutes.

Please pour the iodine solution into the waste cup first.

- (3) Carefully add 2 mL of deionized water along the edge of the rubber tray to rinse off the iodine solution, shake it left and right twice, and then remove the waste.

Pour the liquid into the waste container. Please repeat this step 5 times.

- (4) Use deionized water to destain for 5 minutes, drain the water and record the stain upside down.

- (5) Move petri dish B to the grid in Figure 2, raise your hand and ask the teaching assistant to take a picture and record the starch agar in the embryo part.

Diameter of the fading ring in the blue reaction of iodine-starch in the gum.

C. Amylase activity assay:

Take the extracted samples C and D from the seeds of the re-dried group and control group that have been ground and extracted , and perform...

The following experiments:

- (1) Take 0.5 mL of the extract sample from the dry group and the control group and add 3.0 mL of 0.04% starch solution to each.

Prepare a blank control group (using distilled water instead of the extract sample) and perform the same reaction as above for 10 min.

- (2) Take 1 ml of the treated starch solution into a microcentrifuge tube, add 200 μ l of iodine solution and mix thoroughly.

After color development, add 300 μ l of TCA to terminate the reaction and mix thoroughly again.

$\bar{y}$ (Mixed placement method: Press down with your thumb and forefinger and shake left and right 5 times, as shown in Figure 3)

- (3) Centrifuge (11000 rpm, 5 minutes), collect the supernatant, and then measure the absorbance at OD490 using a spectrophotometer.

Record this. (Please use the centrifuge and spectrophotometer in the common area and follow the teaching assistant's instructions.)

- (4) After step 2, take 1ml*2 of each sample for double-layering.

- (5) Record the OD490 absorbance of the spectrophotometer and calculate the starch content based on the change in absorbance per unit time.

Enzyme activity ($\bar{y}$ -Amylase activity).

Calculate $\bar{y}$ -Amylase activity ($\bar{y}$ OD490/min)

= (OD490 absorbance of blank group - OD490 absorbance of dry group or control group) / reaction time

Please note the correct way to hold the colorimeter tube.



Figure 3

III. Experimental Results: (Total score: **20 points**)

Respiratory activity test (5 points)	
control group	Kaikan Group
Please describe	Please describe

Starch degradation capacity test (5 points)											
control group						Kaikan Group					
Particle 1		2	3	4	5 particles 1		2	3	4	5	
long:						long:					
Width:						Width:					

Amylase activity test results (10 points)						
Test tube number	1	2	3	4	5	6
	One repeat			Double repetition		
	Blank Group-1	Drying Group-1	Control Group-1	Blank Group-2	Drying Group-2	Control Group-2
OD490 nm						
amylase activity γ-Amylase activity						

Please attach the original data to the bottom using tape.

IV. Please answer the following questions: (Total score: 30 points, 10 points for each question)

1. How does the soaking and drying treatment affect seed vigor? What is the empirical evidence?

2. Xiaoming conducted the same study on Mindou, but the results were almost opposite. What do you think might be the reason?

What do you think are the possible agricultural applications of seed soaking and drying treatment? Why?

what?

3. After completing the above experiment, Xiaoming had to pause for three weeks to prepare for an exam before returning to continue the science fair experiment. He then...

The previously soaked mung bean seeds were used to verify the effect of the previous soaking and drying treatment on seed vigor. Results

If you find that the seeds are almost completely lifeless, what do you think is the reason?

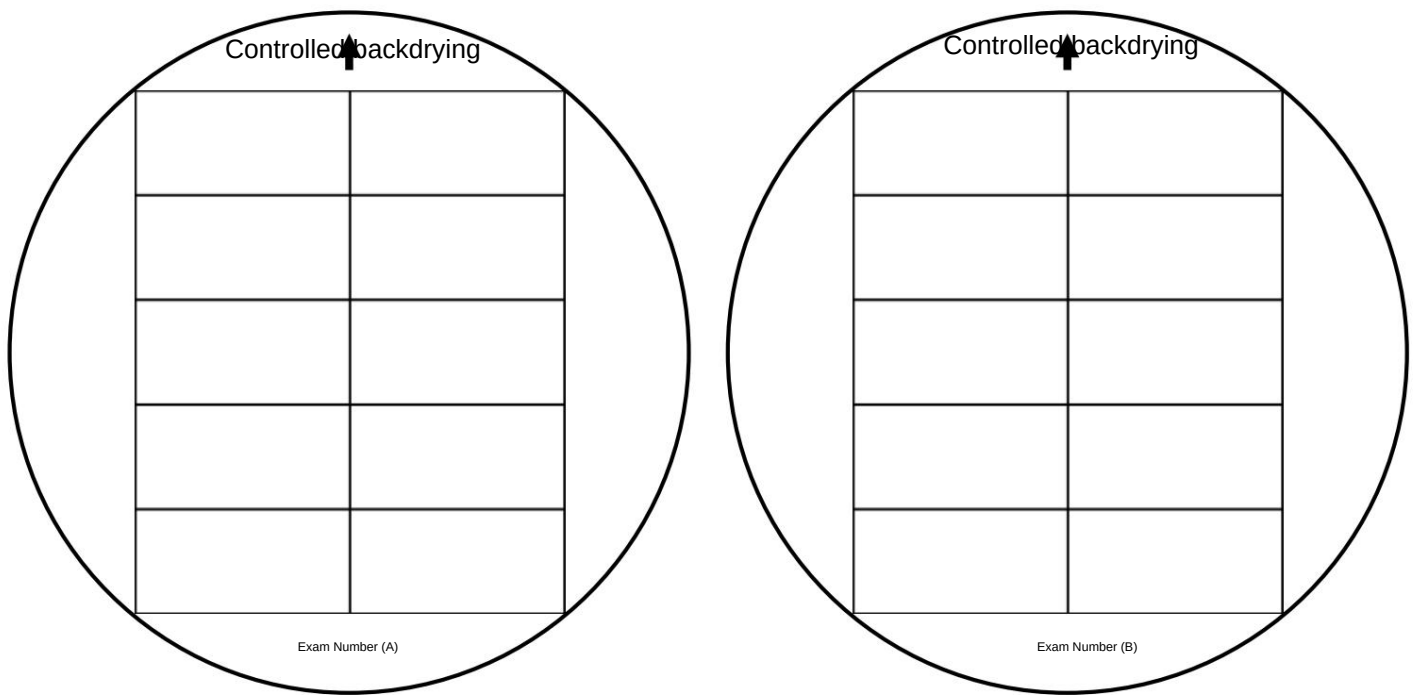


Figure 2. Place petri dishes A and B as shown in the figure. Add the halved mung beans in the order of the experimental steps. Place half of the same seed in petri dish A and half in petri dish B. Please label the lid with the middle dividing line, control group, and rehydration group.

Candidate Number _____

Fraction _____

2019 Republic of China Biology Olympiad National Team Selection Camp Practical Test Questions

Practice in the **4th** Examination Room **B**

Botany practical questions

Materials and equipment:

equipment	quantity
1. Figure 1 (B4 paper)	5 sets
2. Figure 2 (B4 paper)	3 sets
3. Computer	1 unit

Please note:

1. The materials and equipment on the table will not be replenished once they are used up.
2. This exam paper (including the cover and the exam paper) consists of 9 pages and must be returned in its entirety when submitting the exam.
3. This exam includes Practice A and Practice B, with a total time limit of 90 minutes.
4. Please answer the questions on this paper. You may write your answers on the back of the paper, but please indicate the question number.

Ashin, a plant ecologist, was shipwrecked while on a boat trip and drifted alone to a tropical volcanic island. While waiting for rescue, he became so bored that he surveyed all the trees on the island, recording the species and location of each tree, as shown in Figure 1. The distribution map of different tree species is shown in Figure 2. The island is a square island of 25 hectares (500 m × 500 m) with a total of 10 tree species. Ashin used different letters to represent these 10 tree species and counted the number of each tree on the island (Figure 3). The most common tree species on the island is A, with more than 300 plants, while the least common and rarest tree species are I and J, with only 4 trees each.

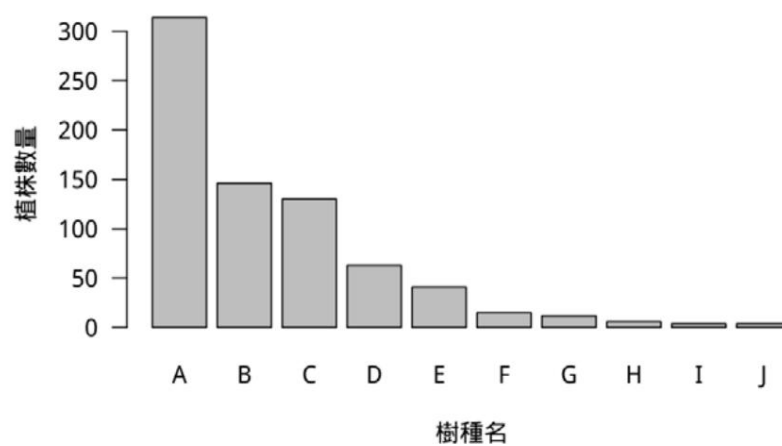


Figure 3. Statistics on the number of 10 tree species on Ashin's island.

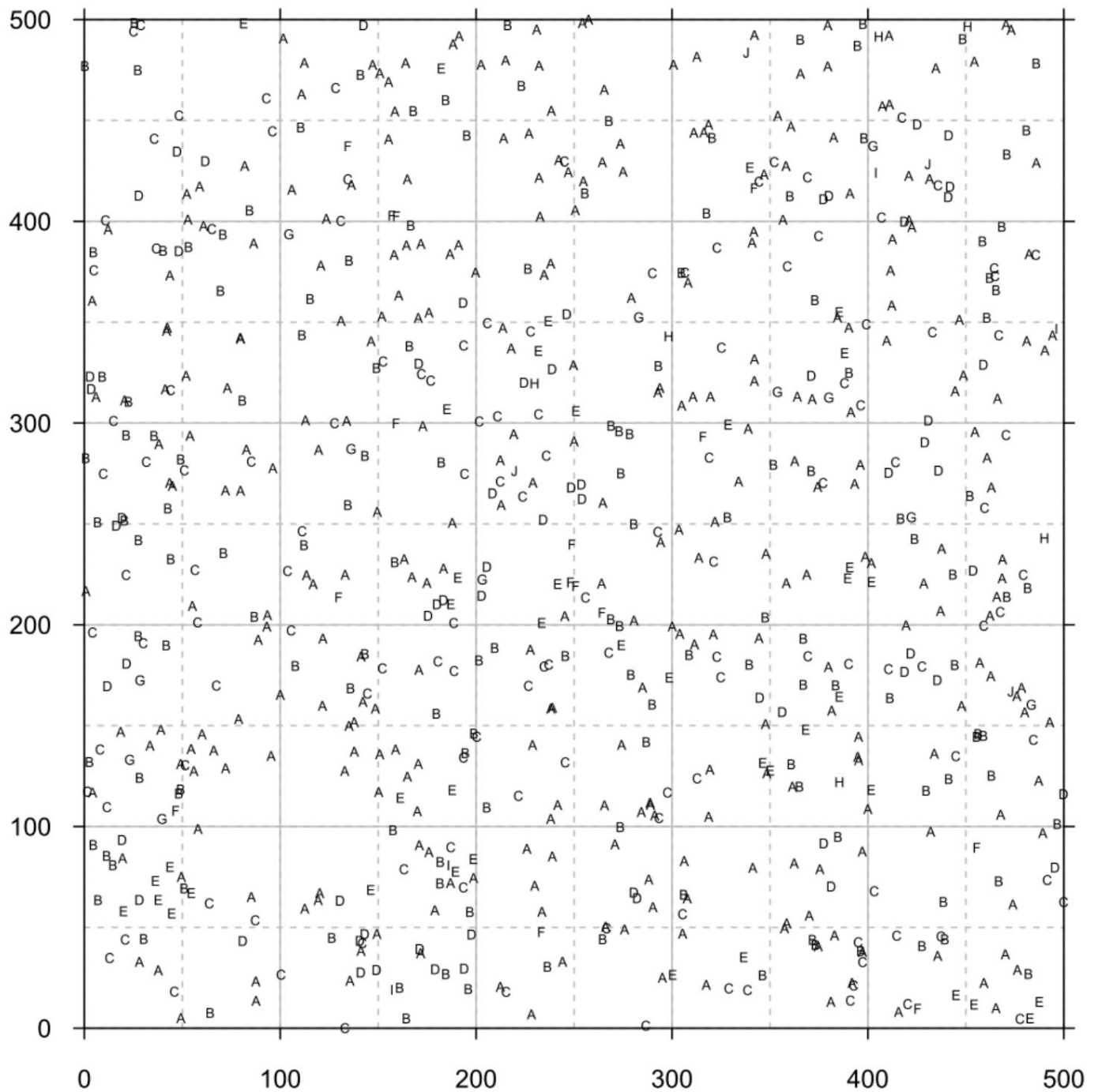


Figure 1 shows the location and species of every tree Ashin surveyed on the island, with different letters indicating different tree species. This island...

The island measures 500 m × 500 m.

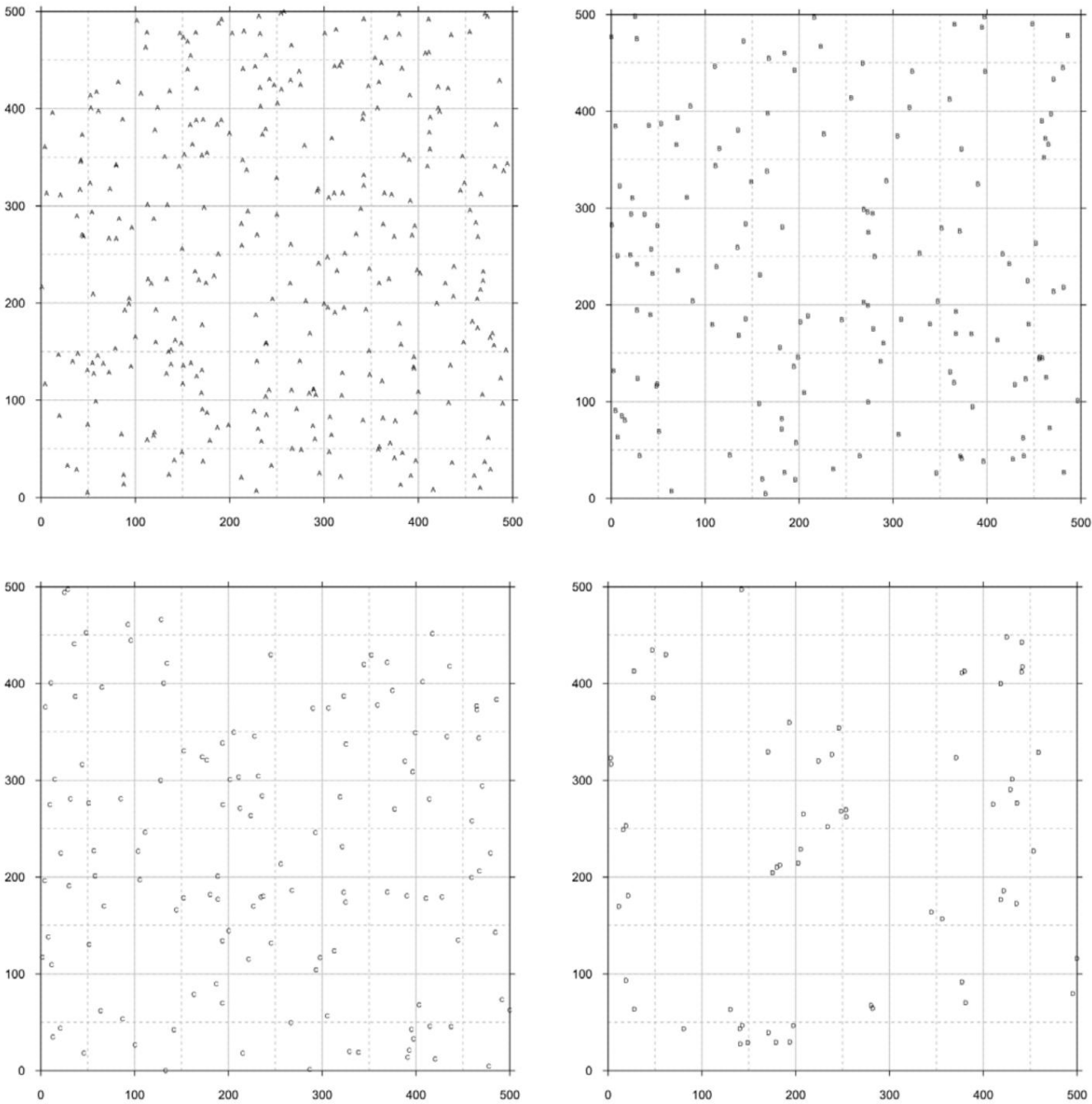


Figure 2. Spatial distribution of different tree species on the island where Ashin is located. The island is 500 m × 500 m in size.

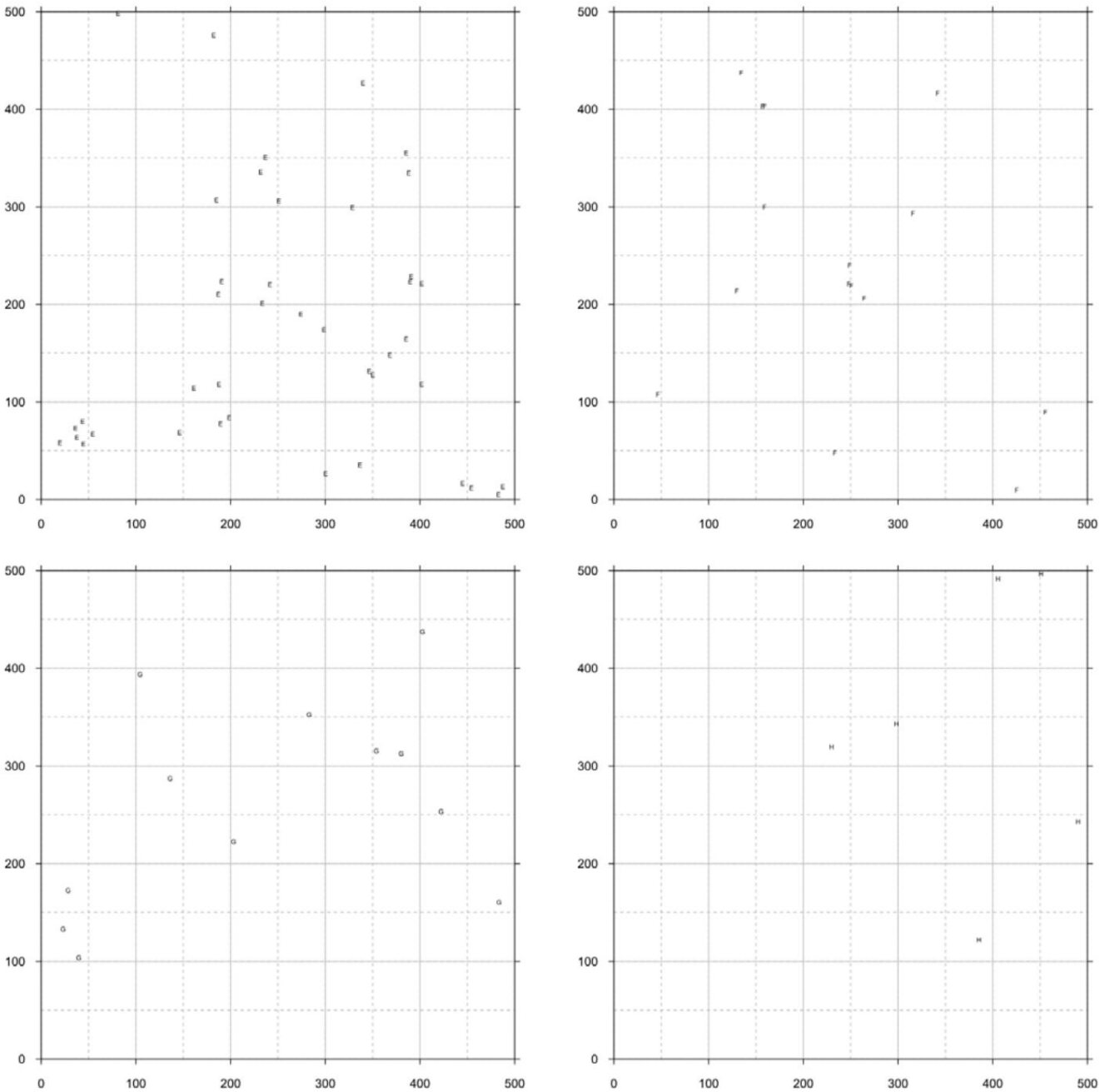


Figure 2 (continued): Spatial distribution of different tree species on the island where Ashin is located. The island is 500 m × 500 m in size.

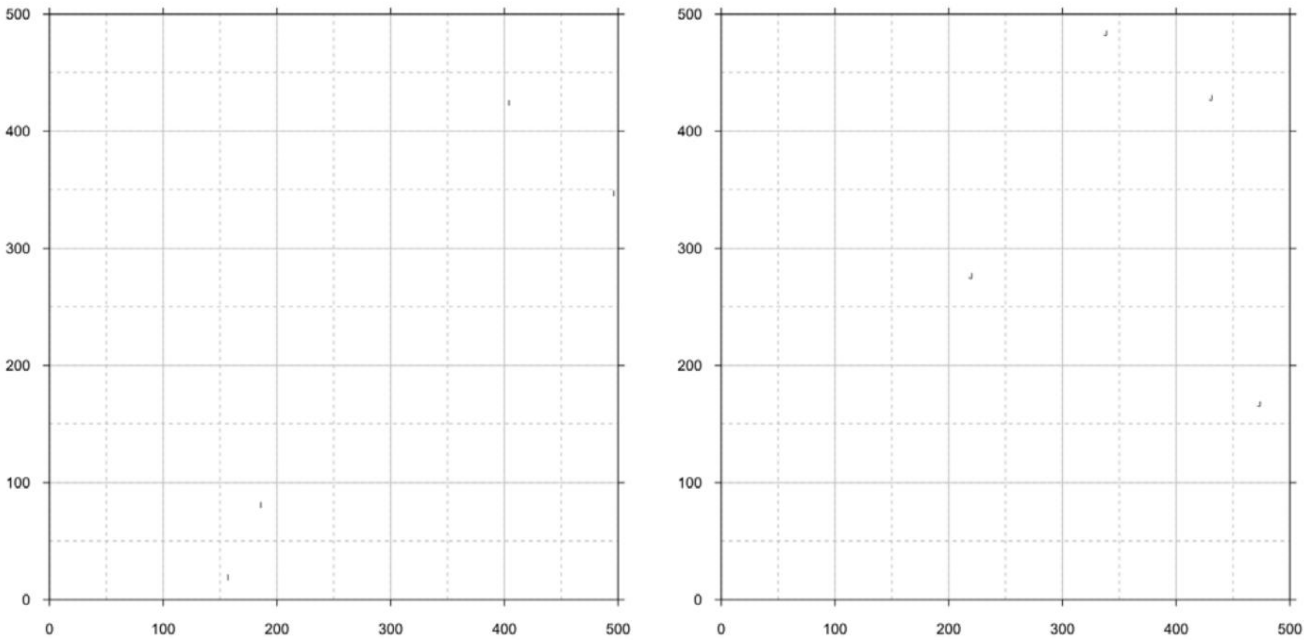


Figure 2 (continued): Spatial distribution of different tree species on the island where Ashin is located. The island is 500 m × 500 m in size.

1. When Ashin was investigating the trees on the island, he discovered that as his investigation area expanded, the types of trees he found also increased compared to the area he surveyed.

There appears to be a relationship. To understand the relationship between the sampling area and the number of tree species surveyed, please use the following different...

The grid size represents the number of tree species contained within each grid cell. The grid sizes are as follows:

50 m × 50 m (0.25 hectares)

100 m × 100 m (1 hectare)

200 m × 200 m (4 hectares)

300 m × 300 m (9 hectares)

400 m × 400 m (16 hectares)

500 m × 500 m (25 hectares)

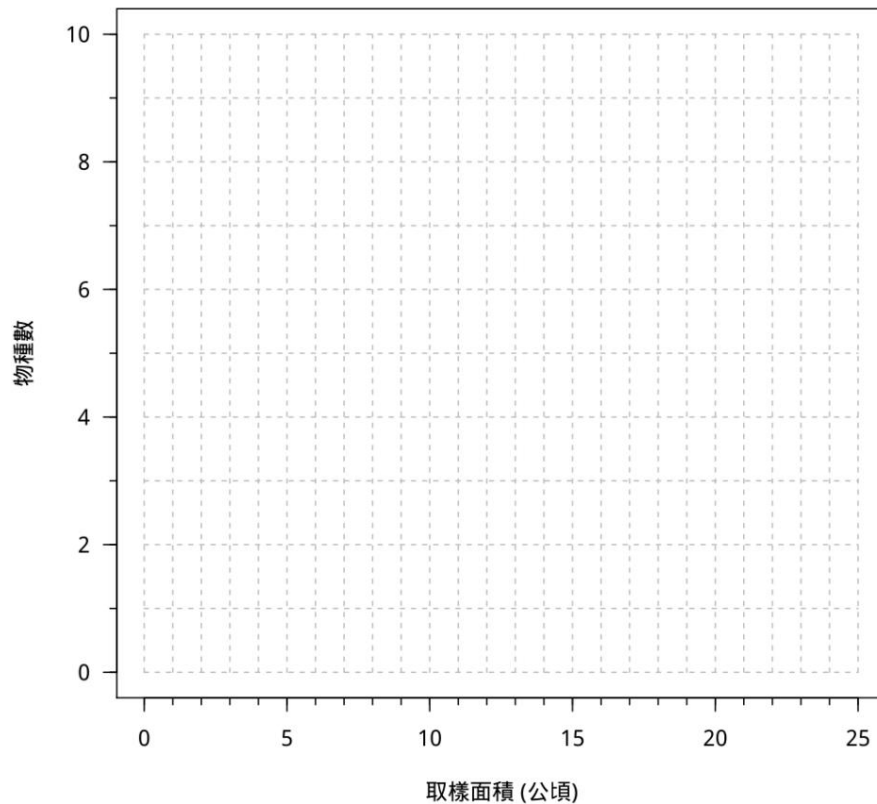
When sampling, for each square size, randomly sample 10 times in Figure 1. When sampling squares of the same size, sample 10 times in different numbers of times.

The sampling grids may partially overlap (400 m × 400 m should be sampled 5 times, and 500 m × 500 m should be sampled once). Please list...

Draw different grid sizes, record the number of tree species obtained in each sampling, and based on this data, plot the sampling area (unit: ...) in the following figure.

A graph showing the relationship between hectares and the number of species (1 hectare = 10,000 square meters). (20%)

Sampling area 1		2	3	4	5	6	7	8	9	10	Mean
50 m × 50 m											
100 m × 100 m											
200 m × 200 m											
300 m × 300 m											
400 m × 400 m											
500 m × 500 m											



2. The species-area relationship is a common method used by modern conservation biologists to estimate species extinction rates.

The method is as follows: If the volcano on the island where Ashin is located were to erupt, and 12 hectares of land were covered by lava, based on the drawing in the first question...

Given the relationship between the number of species and the sampling area, how many plant species might disappear from the island as a result? Please briefly explain your calculations.

Approximately (5%)

3. After understanding the relationship between sampling area and the number of species, Ashin one day got the urge to recalculate the number of species within these squares of different sizes.

The number of tree species appeared, but this time, when calculating the number of species in each square, Ashin used a different method: the number of species in that square must be...

Only individuals of that tree species on this island are included in the calculation. If there are other individuals of that tree species outside this square, the calculation is not performed.

If an individual tree species exists, then that tree species is not included in the calculation. Please see the table below for Ashin's calculation results. Please fill in the blanks using Ashin's new method.

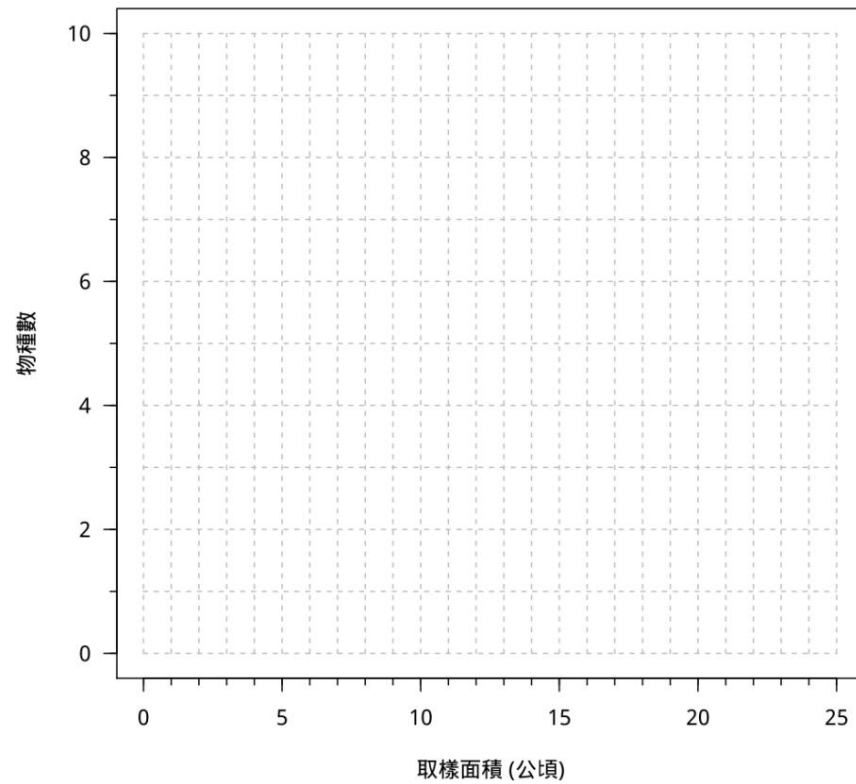
Fill in the following grid sizes (50 m × 50 m, 100 m × 100 m, 200 m × 200 m, 300 m × 300 m, 400 m × 400 m).

The species counts are shown in the table. When calculating, except for the 400 m × 400 m square, please sample each of the other square sizes 10 times, taking [a certain number of samples] each time.

During sampling, the squares of different samplings can partially overlap, and the average of the results of 10 samplings is filled into the table (5 samplings are sufficient for a 400 m × 400 m sample).

Grid size	Sampling area	Number of species
	(hectares)	
50 m × 50 m	0.25	
100 m × 100 m	1.00	
200 m × 200 m	4.00	
300 m × 300 m	9.00	
400 m × 400 m	16.00	
475 m × 475 m	22.56	1.9
495 m × 495 m	24.50	4.9
499 m × 499 m	24.90	6.7
499.9 m × 499.9 m	24.99	9.8

Ah-hsin stated that the relationship between the number of species and the sampling area calculated using this method is called the endemics-area relationship. Please plot the data from the table above in the figure below. (10%)



4. After completing the calculation in question 3, A-Xin discovered that if this relationship is used to estimate the number of species disappearing due to habitat destruction, the results are significantly different from those obtained using the species-area relationship calculated in question 1. Why is there this difference? (7%) When estimating the rate of species extinction, what other factors should be considered for a more ideal approach? Please list at least two different influencing factors. (8%)